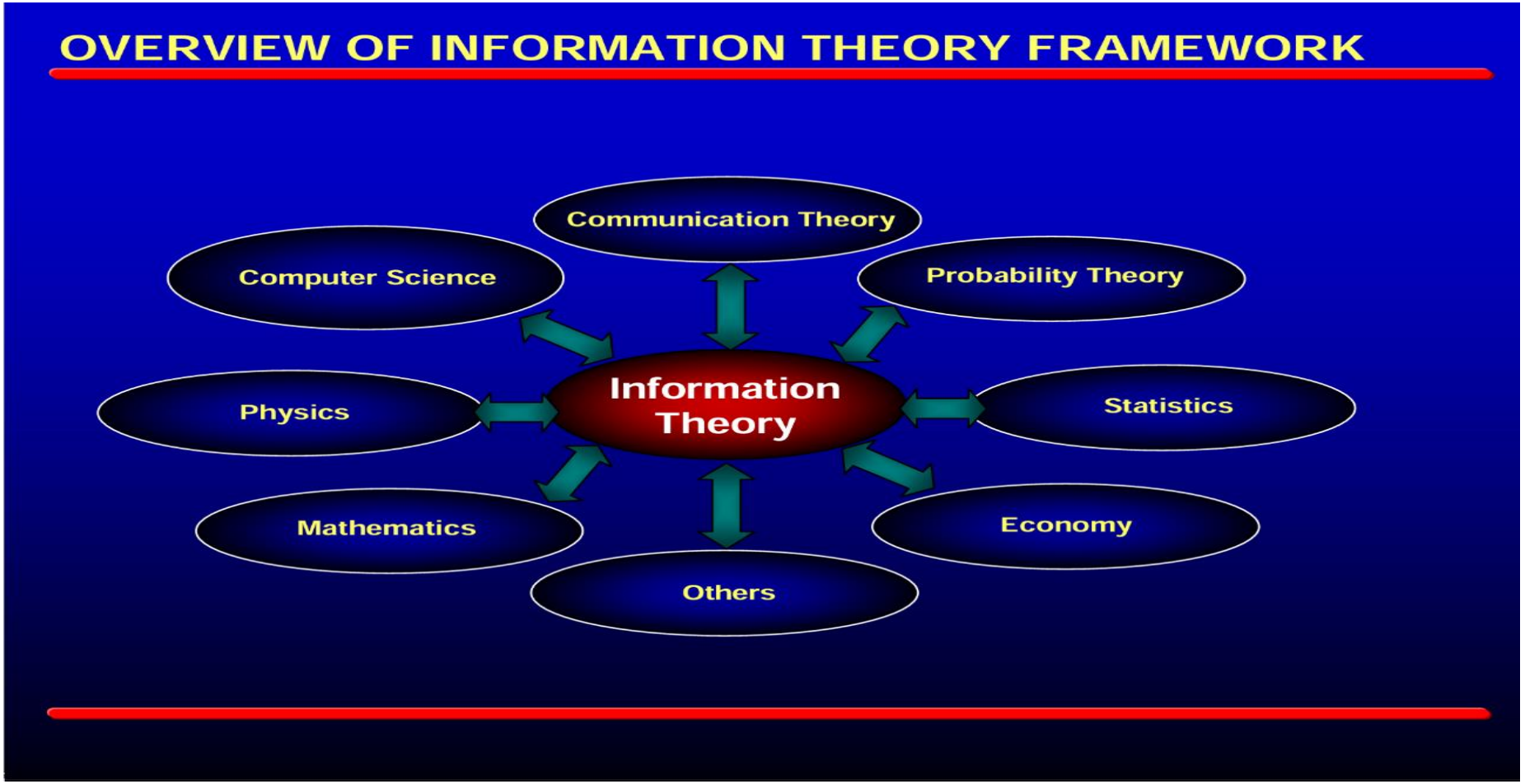
Basic Information Theory

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Information Theory



What is information?

Can we measure information?

Consider the two following sentences:

1. There is a traffic jam on I5
2. There is a traffic jam on I5 near Exit 234

Sentence 2 seems to have more information than that of sentence 1. From the semantic viewpoint, sentence 2 provides more useful information.

What is information?

Let's attempt at a different definition of information.

- How about counting the number of letters in the two sentences:

1. There is a traffic jam on I5 (22 letters)
2. There is a traffic jam on I5 near Exit 234 (33 letters)

Definitely something we can measure and compare!

What is information?

It is hard to measure the “semantic” information!

Consider the following two sentences

1. There is a traffic jam on I5 near Exit 160
2. There is a traffic jam on I5 near Exit 234

It’s not clear whether sentence 1 or 2 would have more information!

What is information?

First attempt to quantify information by Hartley (1928).

- Every symbol of the message has a choice of s possibilities.
- A message of length l, therefore can have s^l distinguishable possibilities.
- Information measure is then the logarithm of s^l

$$I = \log(s^l) = l \log(s)$$

Intuitively, this definition makes sense:

one symbol (letter) has the information of $\log(s)$ then a sentence of length l should have l times more information, i.e. $l \log s$

Information Theory

- A basic idea in information theory is that information can be treated very much like a physical quantity, such as mass or energy.
- Claude Elwood Shannon (April 30, 1916 – February 24, 2001) was an American mathematician, electrical engineer, computer scientist and cryptographer known as the “Father of information theory".

Shannon's Information Theory

Claude Shannon: The Mathematical Theory of Communication

Bell System Technical Journal, 1948

- Shannon's measure of information is the number of bits to represent the amount of uncertainty (randomness) in a data source, and is defined as entropy

$$H = -\sum_{i=1}^{n} p_i \log(p_i)$$

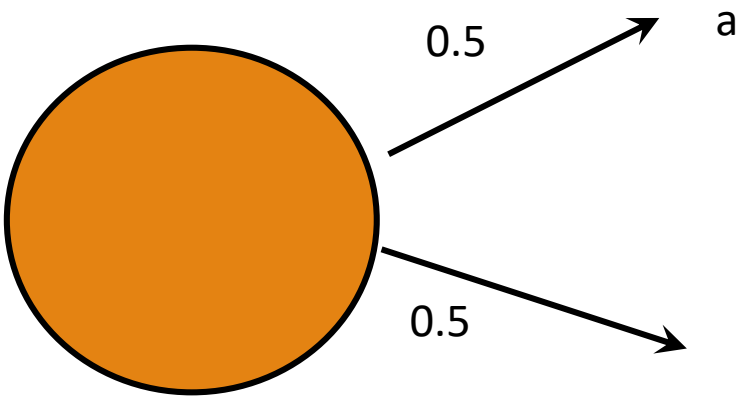
Where there are n symbols 1, 2, … , each with n probability of occurrence of p_i

Shannon's Entropy

Consider the following string consisting of symbols a and b:

abaabaababbbaabbabab… ….

- On average, there are equal number of a and b.
- The string can be considered as an output of a below source with equal probability of outputting symbol a or b:



We want to characterize the average information generated by the source!

Information Theory

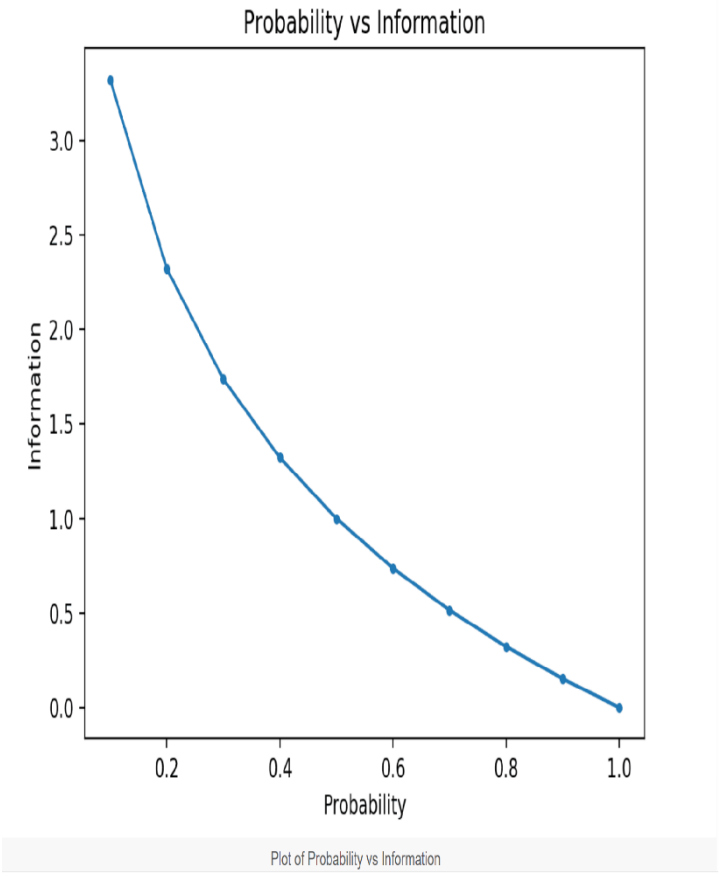
Quantifying information is the foundation of the field of information theory.

The intuition behind quantifying information is the idea of measuring how much surprise there is in an event.

Those events that are rare (low probability) are more surprising and therefore have more information than those events that are common (high probability).

Low Probability Event: High Information (*surprising*).

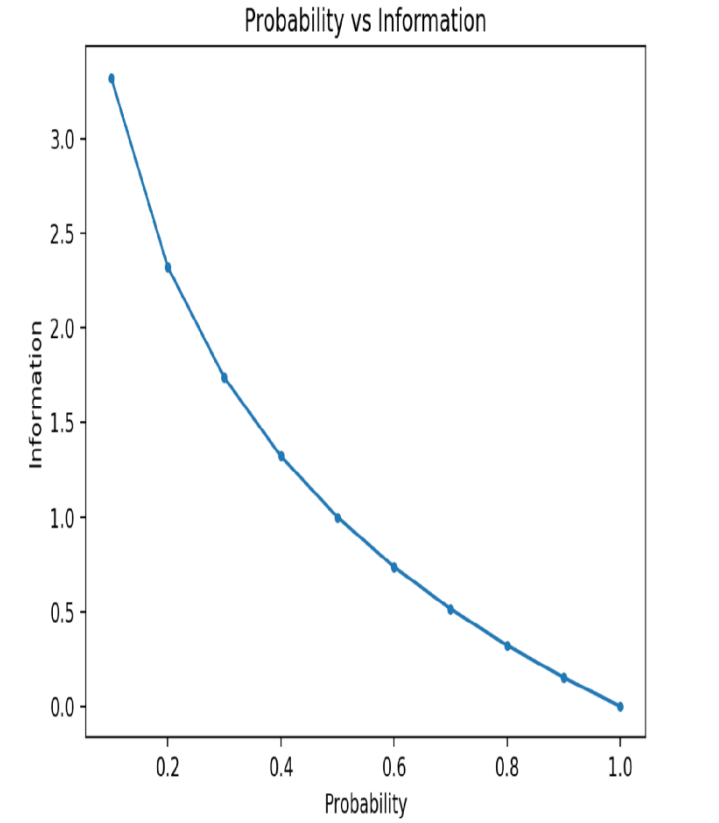
High Probability Event: Low Information (*unsurprising*).



Plot of Probability vs Information

Information Theory

- The calculation of information is often written as *h()*;
- for example:
- **h(x) = -log(p(x))**
- The negative sign ensures that the result is always positive or zero.
- Information will be zero when the probability of an event is 1.0 or a certainty,
- e.g. there is no surprise
- Rare events are more uncertain or more surprising and require more information to represent them than common events.

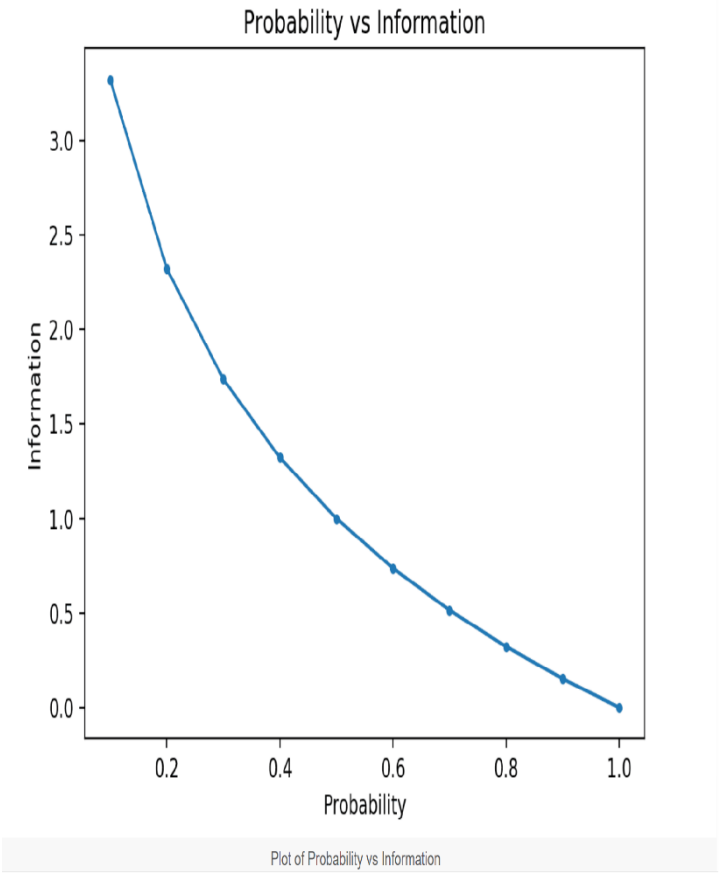


Plot of Probability vs Information

Information Theory

- To calculate information entropy, you need to calculate the entropy for each possible event or symbol and then sum them all up.
- To calculate the entropy of a specific event X with probability P(X) you calculate this:
- As an example, let's calculate the entropy of a fair coin.

$$-P(X) * log_2(P(X))$$



Plot of Probability vs Information

Example 1 “flip a fair coin”

- Consider a flip of a single fair coin.
- The probability of heads (and tails) is 0.5.
- We can calculate the information for flipping a head in Python using the log2() function.
- Solution manual:
- Entropy E= $-P(X) * log_2(P(X))$

$$E_{\text{heads}} = -0.5 * log_2(0.5)$$
$$E_{\text{heads}} = -0.5 * -1$$
$$E_{\text{heads}} = 0.5$$

Since tails has the same probability it has the same entropy as heads (which is 0.5) so we add the entropy of heads and tails to get 1 bit of entropy for a fair coin flip. You can alternately say it’s 1 **Shannon of entropy**, and if you use different log bases, there are other unit of measurements. The important take away though is that a fair coin gives 1 bit of entropy.

Example 1 “flip a fair coin”

- Consider a flip of a single fair coin.
- The probability of heads (and tails) is 0.5.
- We can calculate the information for flipping a head in Python using the log2() function.
- Solution by python 3 under Anaconda, Jupiter platform:

```
# calculate the information for a coin flip
from math import log2
# probability of the event
p = 0.5
# calculate information for event
h = -log2(p)
# print the result
print('p(x)= %.3f, information: %.3f bits' % (p, h))
```

The output is:

```
p(x)=0.500, information: 1.000 bits.
```

Example 2

What if the coin isn't fair though? Here's the entropy if there's a 75% chance of heads and a 25% chance of tails.

$$E_{\text{heads}} = -0.75 * log_2(0.75)$$
$$E_{\text{heads}} = 0.311$$
$$E_{\text{tails}} = -0.25 * log_2(0.25)$$
$$E_{\text{tails}} = 0.5$$
$$E = 0.311 + 0.5 = 0.811$$

- So, in this case, each flip of this unfair coin only gives you 0.811 bits of information.
- It kind of makes sense because each flip you are pretty sure it's going to be heads, so when it is heads, it doesn't give you a whole lot of information.

Example 3

What if the coin somehow is incapable of landing tails. Maybe it is just a 2 headed coin.

$$E_{\text{heads}} = -1 * log_2(1)$$
$$E_{\text{heads}} = 0$$
$$E_{\text{tails}} = 0 * log_2(0)$$
$$E_{\text{tails}} = 0$$
$$E = 0 + 0 = 0$$

- Flipping this maximally unfair coin gives us zero bits of information which makes sense. We know that it will always land heads so the coin flip gives us absolutely no new information.

Any questions

Example 4

We can consider a roll of a fair die and calculate the entropy for the variable.

Each outcome has the same probability of 1/6, therefore it is a uniform probability distribution.

We therefore would expect the average information to be the same information for a single event calculated in the previous section.

Example Homework

Entropy can be calculated for a random variable *X* with *k* in *K* discrete states as follows:

H(X) = -sum(each k in K p(k) * log(p(k))

= $-\sum_1^n pi \log 2(pi)$

NB

log2 can be calculated as lan (x)/lan(2)